

AMR Simulation Performance Report

Source CSV: [dynamic_tasks.csv](#)

This report summarises task completion, wait time, lift usage, and resource recommendations derived from the simulation event log.

Executive summary

metric	value
Simulation start	01/01/2026 05:00:00
Simulation finish	01/01/2026 05:11:23
Simulation duration	11:23
AMRs observed	3
Tasks total	6
Tasks completed	6
Tasks failed	0
Total waiting time	0:00
Total lift time	3:02
Average lift utilisation	6.7%
Recommended AMRs	4
Recommended lifts	3

Method

item	detail
Recommended AMRs	Maximum of workload model and 95th percentile task concurrency using target utilisation 85%.
Recommended lifts	Maximum of lift occupancy model and 95th percentile concurrent lift demand using target utilisation 70%.
Lift parsing	Uses segment_type = lift_transfer and parses lift/floor from from_location and to_location.
Idle percentage	Calculated against the full simulation duration for each AMR and each lift.

AMR list

AMR ID	Qty	Payload (kg)	Speed (m/s)	Battery (kWh)	Charge rate (kW)	Recharge threshold (%)	Start SoC %	Start location
AMR-A	18	500.0	1.2	6.5	2.2	20.0	100.0	AMR-CENTRE
AMR-B	4	500.0	1.2	10.0	1.8	20.0	100.0	AMR-CENTRE

AMR fleet summary

All times in the format of mm:ss

AMR ID	Tasks	Completed	Failed	Task time	Wait time	Avg task	Distance (km)	Energy kWh	Recharges
AMR-A-1	2	2	0	9:53	0:00	4:56	0.58	0.0	0
AMR-A-2	2	2	0	10:43	0:00	5:21	0.57	0.0	0
AMR-A-3	2	2	0	11:23	0:00	5:41	0.57	0.0	0
Total	6	6	0	31:59	0:00	5:19	1.7199999999999998	0.0	0

AMR Utilisation, Idle and Wait %

AMR ID	Util %	Idle %	Wait %
AMR-A-1	86.8	13.2	0.0
AMR-A-2	94.1	5.9	0.0
AMR-A-3	100.0	0.0	0.0

AMR recharge summary

No AMR recharge events were identified in the CSV.

Lift usage summary

All times in the format of mm:ss

Lift energy consumption

Lift	Trips	Total lift time	Avg trip	kWh Consumed	Util %	Idle %
Lift-1	2	0:28	0:14	0.0404	4.2	95.8
Lift-2	2	0:28	0:14	0.0355	4.2	95.8
Lift-4	2	0:31	0:15	0.0429	4.6	95.4
Lift-7	6	1:33	0:15	0.1309	13.7	86.3
Total	12	3:02	0:14	0.24969999999999998		

Lift wait schedule

No lift wait events were identified in the CSV.

Payload schedule

Payload	No. transported	Payload kg
Clinical Waste Bin	3	200.0
food_trolley	3	120.0

Task detail grouped by AMR

AMR AMR-A-1

Task	Outcome	From	To	Start	Finish	Duration	Wait
T1	completed	BIN YARD	D1-WASTE	01/01/2026 05:00:00	01/01/2026 05:06:57	6:57	0:00
RETURN-AMR-A-1-1	completed	D1-WASTE	AMR-CENTRE	01/01/2026 05:06:57	01/01/2026 05:09:53	2:56	0:00

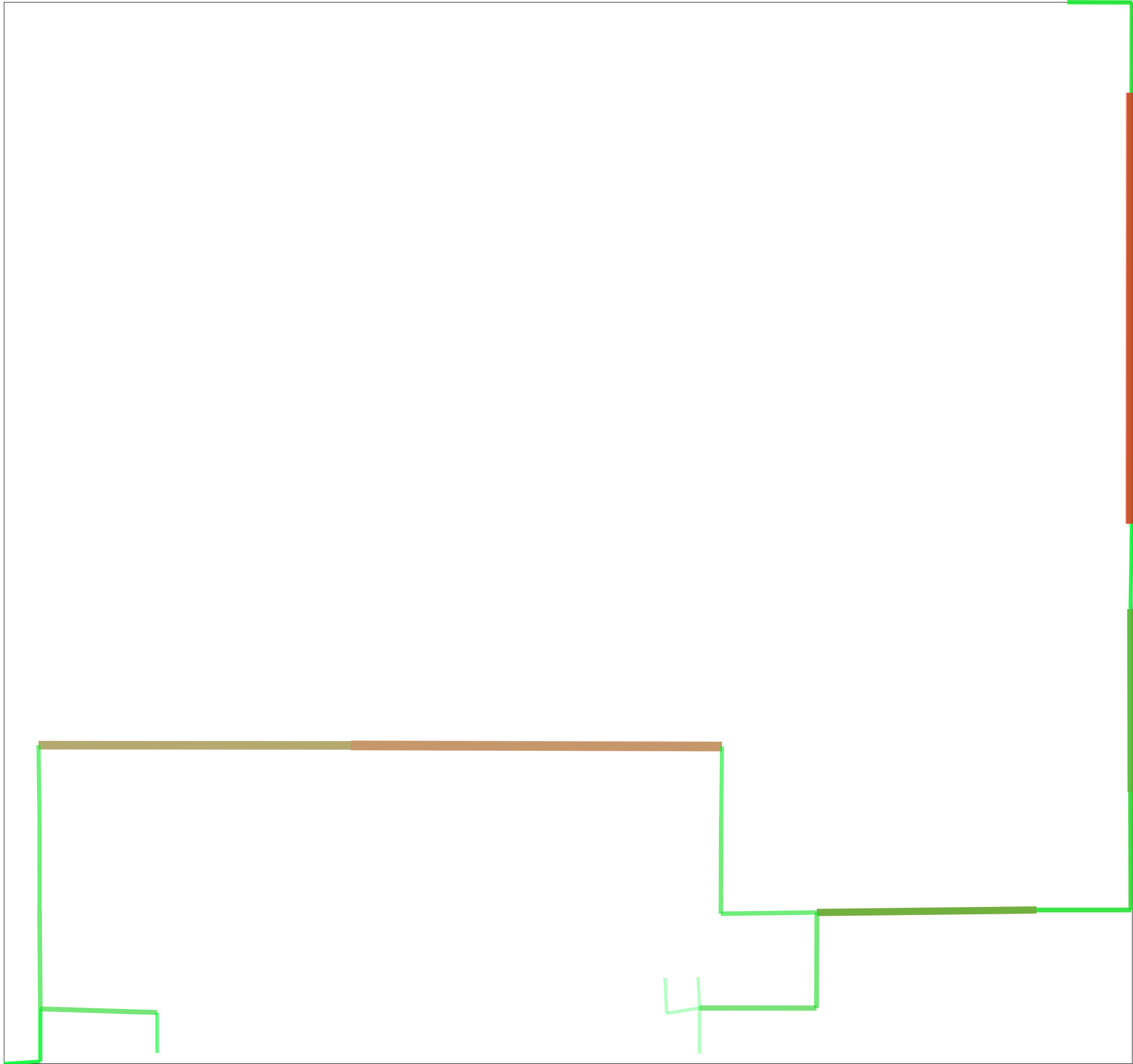
AMR AMR-A-2

Task	Outcome	From	To	Start	Finish	Duration	Wait
T2	completed	AMR-CENTRE	D1-WASTE	01/01/2026 05:00:00	01/01/2026 05:07:53	7:53	0:00
RETURN-AMR-A-2-2	completed	D1-WASTE	AMR-CENTRE	01/01/2026 05:07:53	01/01/2026 05:10:43	2:50	0:00

AMR AMR-A-3

Task	Outcome	From	To	Start	Finish	Duration	Wait
T3	completed	AMR-CENTRE	D1-WASTE	01/01/2026 05:00:00	01/01/2026 05:08:32	8:32	0:00
RETURN-AMR-A-3-3	completed	D1-WASTE	AMR-CENTRE	01/01/2026 05:08:32	01/01/2026 05:11:23	2:51	0:00

Congestion heatmap - Floor 0



Congestion heatmap - Floor 2

